"Modeling Task Behavior with Active Inference" — Ryan Smith

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all right well you know you guys are stuck with me again for for better or worse so hopefully you guys will find this helpful so I've tried to structure it is kind of like a mixed talk where I'm gonna both walk you through some paradigms and some empirical results from those paradigms and then that are based on modeling and then I'll walk you through putting together the actual models for those tasks you first get to see why it's useful and what the results that can come out of it look like and then hopefully you'll be even more motivated to know how to do it um so so I sent out a few MATLAB example MATLAB scripts and so they are they were definitely in an email that everyone should have gotten if you haven't downloaded them then I'm sorry if there's a way you can still get access to them via the internet or whatever that I hope you can otherwise I'll show everything up on up on here I have MATLAB I have the scripts open and if you do have the scripts open right now don't look at them yet I specifically don't want you to have gone through them before because means you'll have the answers to some things ahead of time where I want you to try to come up with them on the fly as we build stuff so okay um so general a line like I said so I'll give you some task descriptions and some preliminary empirical results from fitting models to behavior from people that have done these tasks and a bunch of different and healthy and a bunch of different clinical populations and there's there's three of them there's this approach avoidance conflict task which is what I'll probably spend most of the time I'm kind of building the model in detail and hope is that that'll give you kind of a thorough sense so then I'll walk you through the next task then we can kind of go through the model or briefly and hopefully it'll just if even if we don't walk through it in as much detail for the latter ones I'm hoping that'll sort of enforce some kind of generalization and so you can see how it's not just how the kind of general abstract structure of these is actually very similar across specific models you go and and so actually the way I've kind of depicted this is a little bit wrong it's for each of these I'm gonna give you some what with different levels of detail and kind of hands-on tutorial where we'll build the models will simulate some outputs I'll make sure you guys have some sense how to interpret the simulation plots which are honestly not all that straightforward always in M papers by Carl & Co and and then I'll actually even show you a little bit about how to estimate parameters so to actually get individual difference parameter estimates for each person who's done the task and so at least you know I'm not I'm not saying you'll walk away with this being able to do absolutely everything without any further help but it should give you a good sense of the process and give you kind of the resources to you know practice and move from here and I'm happy to answer any questions or go through further things after its some other point during the week so oh and then and then additionally it occurred to me near the end that not most of this is much more inference based as opposed to learning based and active learning is certainly a really important part of active inference especially when you're doing learning tasks and so there's a little bit of that 3m bandit that I'll go over with you in a couple of slides mainly to give you guys a sense of how you would do something like reward learning um inactive inference and how you kind of think about it okay it gets cast a little bit differently and then I might also walk you through some of the additional kind of simulation paper that we have a free printout on right now on how to use active influence for structure learning which is more or less just an extension of the way you do active learning but I think it's a little bit more illustrative of how of how learning parameters I'm actually works with these models so we'll see what we get there and depends on how long some of this takes so okay so first the approach avoidance conflict task so so it's approach avoidance conflict and basically this is something that happens it's a big problem in psychopathology especially in emotional disorders is that a lot of times you have to choose to approach a situation when both some bats and some good stuff will happen at the same time or you can avoid the bad stuff and go do something safe but that means you actually sacrifice a lot of the good stuff you would have gotten as well and so oftentimes what happens in the context of anxiety and depression is that you'll kind of actually end up sacrificing a lot of good things a lot of sort of well-being promoting activities in life due to a kind of fear of the inability to handle some of the possible negative outcomes that you'd get as well so the and this is a primary target of a lot of interventions both cognitive behavioral and other and basically most likely therapeutic interventions in one way or another target us and so the and this applies for instance in depression and anxiety and you know it can be things like avoiding social activities because of fear of judgment or embarrassment in substance use disorders it could be like drug taking to avoid withdrawal symptoms things like that we're just avoidance is a big is a big thing and and so people have come up with these these tasks that are supposed to mimic having to sort of deal with that kind of conflict and so I'll tell you about one specific approach avoidance conflict task that we have used and how we how you can build a model of it the data so this this particular study and the other ones I'll talk about - all come from this fairly large data set and we've collected in in Tulsa liber where I am where I work and called the t1000 this specifically from the first 500 participants of the t1000 it's essentially the exploratory half of a larger exploratory confirmatory sort of study approach that we've been taking and and so for this 500 this is just kind of like a quasi consort diagram but more or less what we end up with is 59 healthy controls 261 people that have depression and/or anxiety so this is really meant to be much for kind of trans diagnostic and some kind of our document and sort of and then we also have 159 in substance use group which is largely methamphetamine but there is also some opioid and some other ones in there so it's and I should say these this group also can have a bunch of anxiety and depression in it so it's really not meant to be displaying diagnostically carved-up example so okay so the approach avoidance conflict task basically the way that it works is you're shown a little avatar guy like this and you're on this kind of runway and it has one of nine positions and your job is to choose where you want to move this avatar to now the kind of happy son picture over there that stands in for seeing a kind of mildly positive safe stimulus and whereas the kind of rainy cloud that means that you are going to see a fairly strongly negative stimulus but you'll also win a certain amount of points where this kind of little gauge here tells you how many points and it can be filled either to two points four points or six points and so and the idea is that each one of these different positions carries a different probability of getting that outcome or that outcome so basically if you're all the way to the right that means it's 90 percent chance you'll get this guy ten percent that you'll get that guy 80 70 60 50 40 etc so the more you're over here the higher the chance you're gonna get the safe some thing more over here the higher the chance you're gonna get the rainy cloud endpoints thing but it's not fully deterministic visibly that's one of the negative stimuli that you'll get if you uh if you go for the points so point being these are legit aversive you know to the to the degree that you can get through IRB so so you know it's not we turned our best to make it so it's not just these purely kind of you know my negative you know you can rate it but you don't still have that much of a reaction etc but then you get this nice happy you made points right so so there hence the conflict um so but say instead so I should say it's a different trial they're different trial types different conditions where you know the rainy cloud over here Sun over here you know zero points or two points etc I'll show you the different conditions in a second but so say instead you get try to go for the safe thing then there's your so there's your happy safe though and this is really just meant to be kind of like an anchor right like an anchor kind of safe you go to it if you are too afraid of the negative thing to get the points more or less right you made zero points okay so here's the five different task conditions in this case so one we just call avoid threat basically there's just rainy cloud and Sun there's no points involved it's a really you ought to just just all the way over here every time um in this case it's the approach reward condition there's both full Suns on both sides but you get points over here so again it should just be move all the way the right please you're doing and then there's conflict two points conflict four and conflict six in which case it's either the Sun or rainy cloud plus certain amount of points um again two four six so those are the different conditions so the again we're gonna walk through this in a lot of detail but the way that we set up the model for this the exact same structure as what I showed you guys yesterday hopefully some of you still remember a little but city it is you have some prior preferences up here the C thing and there are two parameters that we fit for this essentially it's the value that the the agent assigns to the negative stimuli so essentially how much it dislikes the negative stimulant and then this VP thing which is the value at aside the subjective value assigned to each point you could win and so and so these two things essentially control ought to govern a fair amount of behavior because the more you dislike the negative thing the less points are worth the more conflict you ought to have any ought to be driven to you know toward the Sun but then there's beta so this prior on expected policy precision and which for the paper we've just kind of in also labeled decision uncertainty just to kind of make it more intuitive to people who aren't active inference experts and so we're fitting that as well so it's a measure essentially of uncertainty or or the lack of confidence in your on an action model and in this case because we're not building in any prior habits so II is just flat then this more or less has the effect of making behavior just look more random just less deterministic it would be different if they had strong habits but since since E is flat then it basically just governs deterministic versus random sort of less stable behavior in a sense and the states here are more or less the agent has to have beliefs about where it is on the runway and the agent also has to know things about what task condition it's in and policies are pretty straightforward you can choose to move to one of any of the places and nine places on the runway and the actual observations right so what the agent needs to observe to figure out the runway position in the task condition and to make a decision so there's three different sets of observations one is just runway position cues and the other is the task condition cues so literally just what you see on the screen and then three are gonna be observing the outcome stimuli so the the negative and positive images and sounds and points so again pretty simple and just give you a sense of what the what the a matrices look like so how you'd actually set this up and so this is the essentially the probability of the outcomes given the different positions you could be in and when the trial type or the stimulus or the the trial condition is a particular condition so in this case it would be the avoid threat condition so basically each here is one of the different possible positions on the runway and each row is one of the different combinations of outcomes they outcome stimuli you could observe I'm so under the avoid threat condition essentially the farther you are to the left the higher the probability is that you'll see the negative stimulus and the farther you are to the right the higher the probabilities to see the positive stimulus but the idea is that these probability distributions differ depending on which trial type you're in right so in in in when the trial type is 6 then this joint distribution ends up looking like the higher the more you are over here the higher probability is that you'll see the positive stimulus but the higher more you are the more you're over here to the right the higher the probability that you'll get negative stimulus plus 6 points sorry conditions right here oh oh this yes right so essentially the different observations that you can make the way we set it up are just Sun just clouds Sun plus 2 points clouds plus 2 points clouds plus 4 points in cloud posix place okay and then use that scaling factors essentially that govern how much the value is the value of those outcomes are determined by how much you just like that negatives didn't listen how much each each point is worth to me so anyway so yeah and again we'll go through the model in detail yes the rows are the different observations that you can get right so you can observe the negative stimulus you can observe the positive stimulus you can observe negative stimulus plus 4 points or positive stimulus those two points you know I'm not showing all of them on every one just because it's cluttery but across all of them it's just negative positive positive plus 2 negative plus 2 and I get a four negative six Oh sorry sorry I didn't explain it so that [Music] just for convenience in they just turns I mean this is a bunch of different ways you could you could choose to model this task um the easiest way to do it here is to just give the agent a kind of starting position that's not sort of it doesn't represent any kind of committed position that you've chosen on the runway and so the the first column is just kind of this starting position before you've chosen one of the nine positions and the first row then is just the essentially it's like blank observation before you've observed where you put the avatar so again it's it's just this just for convenience and sorry I should explain up so then one thing that we did so one thing that's important to do a lot of times if you want to try to convince people that your parameters do your estimating in a tasks are actually tracking something interesting or something useful or something real is to try to validate it against some other measures do you think are also going to be relevant to the constructs that you're trying to tap into um and so in this case you know one way to do that is to ask a bunch of questions about self-reported experience or self-reported behavior on the task and so in this case we just asked them a bunch of things you know like I found the positive pictures enjoyable data one two seven how much do you agree or disagree I found it difficult to decide which outcome I wanted I always tried to move always towards the outcome with the largest reward the negative pictures made me feel anxious or uncomfortable this one this one about how anxious or uncomfortable you feel will be a lot of interest to me later theoretically and then also you can measure other kind of standard behavioral things like reaction times and things like that that don't technically go into the model so if the model can say something about reaction times and they didn't go into the model then that can also be valid a-and so just to give you a sense of what this ends up looking like in terms of the parameter validation these are just Pearson correlations for between each of parameters and all those different questions we asked and also with average reaction times and we actually got you know pretty much what you would hope for in a lot of cases so the more you value subjectively valued the points the stronger your self-reported drive to approach with reward was the less difficulty deciding you had the less you avoided punishment the faster your reaction times were etc whereas beta the more kind of uncertain the more like a priori unconfident you were in your action model and so you can think of that again as a decision uncertainty in this case was positively associated with reaction time so the more uncertain you are the longer it took you to make a choice and there's a bunch of these actually they're interesting right like the more you chose to use certain negative emotion regulation strategies for example and probably most interestingly to me is this guy which you have to remember at a sample size of like 500 is actually super significant is that it wasn't actually the negative image value or this value in this case that said much about how anxious or uncomfortable they self-reported feeling it was this decision uncertainty parameter so so what's essentially what this kind of suggests is that in active inference the thing that amounts to or contributes to how subjectively unpleasant the valence of your status probably has a lot more to do with how confident you subjectively think you are in your action model which again is just theoretically interesting yeah this is based on data yeah this is based on the data into 500 people and so it's just a parameter that you estimate so in this case it's largely going to correspond to something like inconsistency in choices across trials in the same condition but kind of it will also be in a way that's disentangled from things about say how much you act as though you value the points and dislike the negative stimulus right so it's it's kind of a way of another thing about modeling in this case is right you can kind of separate out the estimates of the most likely contrib relative contribution of these different sort of independent psychological factors so we'll go through the estimation though but it's just an element in the parameter that you tweak until the model generates behavior that looks the most like the actual participants behavior and and so then here's some of the interesting group differences that we find so when you just break people up into the healthy controls the people with depression or anxiety that don't have substance use and the people with substance use most of which have either depression or anxiety and what it turns out is that actually both of the clinical groups show greater negative value assigned to the negative images so they they're more sensitive essentially they act as though they expect the negative stimuli to be more aversive and in the substance use group they actually show lower beta values which means that they are less uncertain so they act more confident than the healthy people about what what they think the right thing to do is so in this case they're more avoidant and substance use people are more confident that the avoidance strategy is right and so it's interesting there was actually no difference in the point value subjective point value and we also found a difference in prior policy precision between males and females to where females actually looked like they had less confidence in their action model than men on average they're more uncertain or you can call that less impulsive too you know like depends how you want to spin it so then so that's kind of just a you know group difference kind of thing but obviously that's a little more interesting is whether you can use this kind of thing to say something interesting about treatment outcomes and so here what we did is this is an eye I should I didn't I did analyzed the data and I built the model but this is data that's being collected largely by a colleague of mine named Robin a parolee who's a clinical psychologists at LIBOR and basically this this study involves a hundred people that had recruited with generalized anxiety disorder and they were then randomized into two groups and either and one got 10 weeks of exposure therapy and the other got 10 weeks of behavioral activation a bunch of these people also had depression as well as other anxiety related disorders and so again somewhat heterogeneous group and as you can see here response to therapy was also quite heterogeneous and but on average both therapies worked and so we were just interested do each do any of these 3 parameters that we estimated at baseline and can they predict anything about treatment response and pretty cool what we found is is that it actually predicts outcomes for both groups so more or less the the higher your beta value is so the more uncertain you are in your action model the better your response was to both therapies so you measure this thing at baseline then they go through the therapy and it turns out you can sort of say in advance fairly well who's gonna do better it doesn't differentiate responders to one treatment versus the other which is one thing we were kinda hoping for but it does say you know is there B gonna work for you in general relatively speaking and you know I mean my best way making sense of this and I should say this this is for this is based on the promis depression scale and there was a similar result for the promis execs go and so again my best way of making sense of this at the moment is just to say that like look if you're a really confident in your action model and that action model is avoidance then you probably gonna be a lot less malleable right give me less open to trying new behaviors in therapy and being open to sort of counter evidence right so so interestingly the less confident you are in your action model the more we are gonna be to update what your model is about what you should do in therapy I'm is kind of that my current sort of pet interpretation and I should say so this these results we have a preprint up on bio archive about right now or actually it's on if I sell their OSF or whatever that one is but these we don't this we don't have anything out on yet and because we're actually waiting collect more data to finish it up and but it's close to being done and so summary here is that you know it looks like avoidance in depression anxiety and substance use looks like it could be driven by more negative expected outcomes and greater policy precision looks like it may indicate greater confidence in the avoidance strategy in substance use and the less confident you are in your avoidance strategy the the better your you look like you respond to therapy so and I should say that one thing when you're building trying to build a model of something like this that's worth emphasizing is that there is no learning in this model and this is a purely inference based task you know you know what the stimuli are gonna be and you just infer okay you know what's my best bet based on the outcomes I want to observe based on my relative preferences about each outcome and the probabilities and so using something like I kind of standard reinforcement learning model that requires you to be simulating learning doesn't necessarily work so well whereas active inference is kind of nice for this because you can do a purely inference based model um it's not the only model that can do it but it's it's a nice feature of active inference models that you can do that um okay so now that hopefully I've motivated why this can be useful let's let's try to build it and so I showed you guys briefly how I structured the model maybe you've forgotten by now but so kind of just walking through the logic so let's you know part one of the model that I discussed yesterday so just observations

States how states are related observations and what your prior expectations over States are so those are the first four parts right so what so what are the states is the first thing that you need to specify when you're gonna build a model so then the question is what does the agent need to know to do the task so anybody I mean I told you a while ago but if you don't remember any any any thoughts as to what what you would need to know to do it that is one you got to know where you are to be able to know where you want to be you know yes you need to know where you are and yeah the other places that you could be right so position on the runway that goes and it's gonna need to be one hidden state factor um what else not max well no so the points aren't actually something that needs to be hidden state the points are going to be something that you observe right they're gonna be an observation that may or may not be preferred right so so what else what else so the image is also going to be an outcome it's gonna be a thing you observe the probabilities are going to be in the a matrix those are the things that map the states to the observations values of the points is also an observation the subjective value of the points is and that's going to be in the C matrix which we haven't gotten to yes trial-type or TAS condition remember cuz we're you're gonna want to move on the runway is going to depend on knowing what condition you're in right so so those are two things you need to know where you are on the runway and you got to know what condition you're what task condition you're at so minimally a minimal model of this task requires that the agent knows those two things and it's sufficient that it knows those two things yeah um yes and in this case they're fairly trivial it's just what you see on the screen so there's no uncertainty yeah so we'll get to that but I mean it's just an identity matrix it's very yeah any other questions at this point yeah um well you won't know where the Sun is so I mean actually that's that's a good point um well interesting yes I agree that if if no matter what well so that would I mean that would kind of make you go wrong in condition to here right because it's Sun versus Sun plus points and so in this case if you just had the policy of always going toward the Sun then you wouldn't you wouldn't have any idea of what's on the other side sure sure well that's I mean that is about the negative expected you know negative value of the outcomes right ahead of time the governs decision-making and that is dependent on the context right that's dependent on the knowing that they're getting on the train as opposed to being in some other situation yeah I mean so in this case I mean point point totally taken you know in this case I you know when it comes to the automaticity of what actions get selected I mean if remember these I mean I didn't say that but the it's not as though the left and right positions of these are always stable right so I mean they do flip back and forth and in counterbalance ways so it can't just be like I'm always gonna go right I'm always gonna go left or something like that right so I mean if it's that that level of automation that I don't think it's a problem but but I agree you can adopt the just play it safe no matter what sort of strategy which I'll show you some people behave that way so which just falls out of the model so okay so states only position in task condition all right so um so in practice in the actual scripts that we'll walk through in a second and so this is just a little kind of cut and paste from part of the script and in practice you specify the hidden States and the number of levels teach hidden state by specifying the D vectors which are your priors over states and so in this case and there remember there are two time points in the trial right there's the starting point where you make the choice of what position you're going to go to and then is the position you go to and so the D here is your prior over initial states so um so one then how many different levels do we need for hidden state vector 1 which is the positions anybody so - nine is what you would think right but four because of the little a kind of trick that we used sexually nine plus one starting positions starting plus the nine positions you can choose and and what's the prior what's the prior over that gonna be yes exactly yeah exactly wanted a bunch of zeros you're completely confident that you're always going to start in the starting position right that's all that means so how about for e to so how many how many levels are gonna be in that end state vector there are five yes and what will the what's what should the prior look like over that what's the prior distribution gonna look like yeah so in this case yes it'll be flat so so in practice it can just be once I mean this this is just gonna get normalized so does it doesn't matter so so just saying the agent has no prior beliefs that one task condition is more likely than any other yeah you could and and for this is kind of getting the stuff that we'll get into later worth learning for capital D makes no difference because it's just gonna get normalized when it comes to learning priors which will be little D then it matters because they stand in for concentration parameter values but again everybody ignore that for now okay so so then so by setting up that that way you're saying there are ten levels in hidden state vector one and there are five levels in ends de factor two and those are the priors over those levels okay so then we already kind of talked about this a little bit but what does the agent need to see or to observe to know what state sits in right so that's gonna be your observations I already told you but anyone right so it needs to see this thing on the screen right it needs to it needs to see what position it's in anything else exactly so it needs to see these cues because those are at what it's gonna those are what's gonna tell it what task condition it's in right um anything else oh well that's I mean that's just that's just gonna be part of the cue that tells you what condition you're in this is really actually really really obvious what else does it have to observe yeah has to observe the the actual outcome stimuli that it does or does not want to see and hear right so so so yes so then those are so those are the three observation modalities your runaway position cue is your task condition cues and what all right whatever that was okay so so right so runway position cues task condition cues and outcome stimuli all right so the idea then is just to make this clear is that the that those observations those three sets of observations are gonna be generated based on interactions between these two where you are in these two hidden states right so for instance being in that position combined with being in this condition will convey a 90 with 90% probability generate the observation of the negative stimulus and 6-plus right the probability is true the agent the agent knows the probabilities and they're correct yeah so there's no like I said this is purely an inference test there's no wiring okay so in practice when you look at the script the way that we set the number of outcomes per out the modality is just with this kind of bit of code that more or less just sets up the a matrix where all its really doing is it's saying based on D based on the number of levels in each hidden state factor make that many columns in in the a matrix and then for each outcome modality give it the number of rows that correspond to the number of observations for observation modality so in this case it's just going to be so you have ten positions right five conditions because these are just one to one right you observe the position you know the position you observe the condition you know the condition and then for the stimuli what I showed you earlier is the way that we chose to do it is there's seven different formal observations that could be made in terms of the signaling go over what oh right so I will when I go through the images so so then the a matrix right so the idea is is that that is explaining how states generate outcomes which you can then invert to use for inference so you observe the outcomes you say conditional on those outcomes what states am I most likely in and so here you need one a matrix per outcome modality so per type of observation that you can get and so in this case we'll need three a matrices one specifying how states generate the outcome stimuli one specifying how the states generate condition cues and one specifying how they state position how they generate position cues and so essentially it's just that expanding out this arrow about how states map to observations and so here is what the first a matrix will look like so the one specifying how states generate outcome stimuli and so remember here so the way we've set it up is there are 1 2 3 4 5 ok yeah all right yes sorry that is my thought so you should have this one circle as well but so so there should be 1 2 3 4 5 6 different outcomes you can observe plus the starting position 1 right so that corresponds to 7 rows in your a matrix and so the columns then are each of the different positions you can be in one being that starting position and then each row is going to be one of the different observation one of their from observations you could make and then the entries here and code the probabilities of observing each outcome given which state you're in but remember these distributions are conditional on what condition you're in right so that's what this kind of third the third dimension of this a matrix is is saying conditional on being in the you know this condition you know the probabilities look like that conditional on being in condition 2 here which is this guy the probabilities look like that etc so it's it's encoding these probabilities again which are known and then the other ones are really just kind of trivial right so you can either observe the cues that you are in the avoid threat the approach reward the conflict to for insects and the states here remember are so are you know this one stands for being in condition one condition two condition three etc so this is just an identity mapping along the third dimension essentially so if you observe these cues then no matter what position on the runway you're in you're in vehicle you believe you're gonna believe you're in the avoid avoid threat condition that that all make sense if anyone is confused by anything again this is like for you guys to learn so we can take like what are way as long as we need with us very very it's just saying what you observe tells you with a hundred percent probability will stay here at so there's no basically there's no uncertainty at all and in what you observe tells you about what's going on in the task yes no yeah the partial partially observable aspect of this doesn't matter at all for the aspects of the tasks that don't involve any uncertainty right so in this case in this case it it's necessary because of the probabilistic reward and Punishment outcomes right and the M I mean yes exactly because that's for your specifying probabilities right and equally trivial for position queues and again it's just it's just saying it's night any matrix for each of them so just ones down that I a get all position equals position right so again totally trivial but that's the true psychological ground truth right when you see the you know where the avatar is you know where the avatar is okay so any questions about any of that because the next thing we'll do is we'll add in the temporal dimension chirp-chirp yeah formerly it's very easy to do you just make these things not once the so they did what the task condition is mm-hmm mmm miss reading the context cues yeah absolutely that's actually really interesting yeah I mean the context of this task in particular it's probably yeah you'd have to design a different task where the context cues are a little more ambiguous but absolutely yeah and then formally yeah you just instead of this just being you know ones you could just you know make it blurry or and yeah I mean you'll say I mean obviously like you know using this again where we're using models that aren't fully sort of Bayesian inference based when making decisions when there's probability is right like and it's still I mean I still thought it'd give us interesting results with the parameters rate but clearly these models are fancy enough that they can handle much more difficulty less problems absolutely okay so then specifying the B matrix here so again conditional on being in state one how you think you know what your prior is for what state two is gonna be or what the state is going to be at time two that then you're gonna need one B matrix for each condemned B matrix per state factor right so basically that's saying how do you run away positions how do you believe runway positions are gonna change over time and how do you believe that conditions are gonna change over time within a trial that should be that needs to be clear because um the agent is just going to be equipped with the belief that the condition is always stable across a single trial right it's not like the queues are gonna change as it moves the avatar but you're gonna need one matrix for the runway position observation way up sir runway position state factor but because this runway position is under the control of the agent they were actually going to need to be nine B matrices for this one B matrix corresponding to each possible action that the agent can choose where an action is here specified as a transition so so this is important that that well I'll get to this in a second but basically what the policies are going to be when the policy PI thing appears up here is just selecting one of a bunch of possible B matrices one of a bunch of possible transitions that are under the control of the agent so again trivially just a little chunk of code here that just says for condition it's an identity matrix you know the the task condition is not going to change while the agents trying to make a decision about where to move whereas this the second part here is just saying that for each of the different levels in hiddens de factor one right so ten there's going to be a bunch of ones for each across one line for every possible action so at the end of the day that just looks like this right so action one so the B matrix one here action one just basically says no matter what position you're in if you choose this B matrix you're gonna transition to state one so column their states and rows or states at t plus what time at the next time point so if you're in one you move the one if you're in - you move the one if you're in three you move to one etc so that's action one action two is no matter which state you're in you move to stay - and then you know dot dot dot until action ten is no matter what stay here and you move to say ten and so each of these is a possible action that can be chosen each of those transitions and in this case policies are going to be identical to actions because there's just one action it's just one step but in practice when you have tasks that require making selecting sort of chains you know sequences of actions which is what a policy is sequence of actions then allowable policies are going to be allowable sort of sequences of these right so it can be like you know be matrix - then B matrix go on to me matrix three and so forth and that's why you need a separate you need to separately specify what the agent is allowed to do in terms of allowable policies because maybe you want to say you know action - 3 - is allowed but not to - - there's something like that right so you have to specify what policies are actually plausible for the agent to pick windows consists of different possible sequences of actions or sequences of transitions that the agent has control over so in this case it will just look like this it will just for the for the second factor so for condition it's just all it's all just going to be ones because there is only one action that can ever be chosen just stay in the condition but for the runway position this is saying that the agent can choose action - or action 3 or action 4 X + 5 or action 6 7 8 9 or 10 there's no one because one would correspond to staying in the starting state which they're not allowed to do and then again a little snippet of code for that mm-hmm yep mm-hmm so there are some cases where what looks like learning can be captured as inference there are many other cases where it can't and the particular case you're talking about I don't think you can capture that purely as in for insulation that way is that popping in mind right now I'm in which case you'd have to activate the learning component of these models which I will talk about a little later but more or less here just and with each observation so on each trial you know whatever belief you or whatever belief you have about what state you're in you essentially tack on a count that says that state is you know that much more likely because I observed myself starting in that state in the past or every time I believe I'm in a certain state and I observe a certain outcome then attack on account in the a matrix that says okay now I believe that that state can generate that outcome is a little stronger and so so over time essentially you're building up priors about what states you're going to start in and what states are gonna generate what observations just through previous exist through repeated experience it's essentially it's it centrally just heavy and just heavy and coincidence learning and just mm-hmm so there are so the reason I said that is because so one way to model that is to have different additional different possible observations where some observations could involve even more negative stimuli or something like that in which case you would learn that hey being in this position actually generates even worse observations in which case that would be a matrix learning the other way you can do it is by somehow involved somehow model learning the actual preference distribution itself which you can do but it's a little bit kind of beyond the scope of this that's what I just yeah yeah you can you can fiddle with C but um learning C is a little bit more complicated and but you can certainly fit C values like we do right to see what C is for each person for each possible observation so so yes yeah I mean hopefully hopefully want to get to some of the learning stuff that'll be a little bit ok so here so preferences here so your your C vectors so I'm just starting out saying you know these are just a bunch of zeros for each possible observation modality and then I'm saying okay so for the first one right the one where I'm observing what the stimuli are that I observe at the end of each trial these have a particular value assigned to them at time point two right so column one is time point one column two is time going to and I'm parameterizing these so here I'm saying that the value of negative image is just this parameter neg underscore IMG and then positive underscore IMG and then this one is the same parameter again but plus two times whatever my subjective point value is and then so on and so forth so there's two parameters I'm fitting here there's those three there's negative image positive image and point value and but as you'll see what we actually end up doing to make this invertible is you just fix the positive image value at 0 this kind of like an anchor value and then the other two get adjusted around that so if I were to choose the parameter value of negative one for image zero for positive image and point value equals subjective point value equals one then that's what the C vector would look like it'd be negative one for the negative image observation two for the positive image plus two one for negative image plus two three and five so this is just saying how much the agent for first one over the other but again I mean this is just arbitrary based on the app based on the values of the parameters that I picked but we're gonna fit those and figure out which one's best reproduce each participant's behavior okay and so then the last part of the model then right as you have yes sorry so so each column is a time point yeah so a time point one it's just always gonna be in the starting position so it doesn't matter their time point to it prefers its preference distribution looks like that or higher equals more positive so then II like I said we're not assuming any kind of habits here so he's just a flat distribution with ones over each possible policy and we have to set some value for for beta so for prior policy precision in this case we're fitting that as a parameter as well so I just put it as an example you just buy beta equals four but but this is something we're fitting so we're gonna get a value for this for each person um and that's really it then you know you've defined well so T is just the number of time points so just T equals two in this case but so you've defined each of these things you just throw them into a structure that's labeled little MDP and these three I've commented out because as I'll go into if you want to if you want to include learning then you also include you can also include a little a little be your little D or a little C or a little e actually but I just in practice don't use this much but these are all commented out because we're not learning isn't happening we're not separating out the the actual generative process from the agents model and and again beta adjust equals MVP beta equals beta here which we said in the example being four there are these other two that are worth mentioning so MVP alpha is a essentially a an inverse temperature parameter it's like you can think about it as just encoding some some level of like behavioral noise or motor stochasticity so I think kind of like shaky hand you know when you're pushing a button there is always a little bit of that in human behavior and then Etta is learning rate um basically governs how much of a count you actually tech you know tack on every time you have a new observation but again that doesn't matter in this case because we're not including learning so once you have all that then you you know say your parameter values for a simulation and then you plug the hole MDP thing into this very very dense and opaque function that carla wrote called the SPM - MVP w or underscore vb underscore x script and it will take your model run it and the the message passing algorithms that minimise free energy and it will spit out the results what the agent shows and so this is an example of the kind of simulation outputs that you'll get there's different plotting scripts depending on what you want to look at but in this case the plotting script for a single trial is this MVP underscore vb underscore trial function and these are all just built into SPM in the de m toolbox so they're all just as long as you have SPM twelve-year it's all there and so the way that you can kind of read this is the darker the color the higher the probability so in this case the stronger the belief of the agent and the little cyan dots represent kind of the ground truth so in this case the agent believed that action 10 so moving to position 9 was the best move to make at the end of the day and that is what it shows this is its beliefs about position so it believes that at time 1 it was in position 1 and believes at time 2 it was a position 10 sir state 10 state state factor 1 level 10 so position 9 and that was true and it believes that it was in the first condition the avoid threat condition so more or less it just moved all the way toward the Sun because that was the that's the thing that made the most sense to do because it was correct about what condition it was in now like like you were saying if Lawrence if if it had some uncertainty about what condition it was in then it wouldn't have been near that confident but were the right place to move with that right so this could have been more kind of gray I got more kind of diffused probability distribution over different states then you'd get different behavior and ignore that that doesn't really mean anything this is just saying how it's this is just the posterior probability over policy so it's how the policies essentially how confidence in policies change over time so this is saying it was before it saw what position and condition it was in it was completely uncertain about which of the nine possible policies it should choose and then after it saw the position and condition it jumped up to being very confident in just the policy of going to the ninth position that update in confidence from from before to after seeing the position and conditioned stimuli corresponds in the neural process theory to getting a little kind of bump in dopamine so that's where that that kind of change in policy precision is what uh what you would simulate as kind of physic dopamine response if you were going to try to use this in like a model based fMRI analysis to see if you could you know throw that into throw that like trial by trial prediction to like say like an fMRI simulation and see if you can actually track things that look like they plausibly correspond to dopamine sending or receiving brain regions for example Phillips quarterbacks actually done that in a paper from a couple years ago and showed really nice ventral tegmental area activation as well as cortical activation and a bunch of known dopaminergic target regions so well I mean basically it just it just like it's just very confident in what the best decision is right so i mean ii ii did because remember in this condition this is the avoid threat condition it's basically there's a bad thing and there's a good thing there's no conflict there's nothing so it's just fully rational like there is no uncertainty at all about you should just move toward the thing that's gonna be good it would be different you know in the different conditions especially if beta was a higher number because it would have a lot more uncertainty so anyway so then here at the bottom this is showing the actual observations and the colors here correspond to the relative preferences and the preference distribution so basically it doesn't care what conditions it's in or doesn't care what position it's in but it does have interesting preference differences over the observations it can observe the outcomes that can observe at time two and so in this case the distribution looks like that and what it actually observed was this guy which I'm pretty sure is just the positive stimulus or something so that is more or less how it works beginning to end you can then you can then run that repeatedly to get multiple trials so in this case for the task in question we have sixty trials I'm in total and so you can run the thing under different parameter values and have the agent do it 60 times in a row then you can get a sense of what the actual behavior across the task would look like under different parameter values and then you can do that a bunch of times and our different parameter values until you find the set of parameter values that best matches a given participants behavior and then those estimated values we'd actually use in further individual difference analyses and the way to do that or at least the way that we've done it is using this variational base function that's also just built into SPM that minimizes variational via variational Bayes it's this SPM underscore and LSI underscore Newton function and and so what you need to do is you just take an agent's observations from the actual task and what the actions they chose work on the task those go into these U and Y variables and and then you have to set some priors for what you think good kind of starting parameter about good plausible starting parameter values might be for anybody and then what it does is it will just start with those priors and then iteratively adjust them around until it finds values that minimize the log likelihood ie maximize free energy ie produce the behavior that's most consistent with the actual agents behavior or the actual participants behavior um but doing it this way when you start with particular prior values this particular algorithm it penalize --is moving the parameter estimates farther from the prior values that you set and that's a way of avoiding overfitting the idea is the farther you need to move parameters from the prior value that you thought was plausible the more kind of the more overfit something might look so by penalizing that you're it's a way of preventing overfitting so it's finding the best fit parameters without moving them too far from the priors you set and so in practice we do that and it will look like this where over time so is it with each iteration it will give you a log evidence or energy and over time what you hope to see is that this goes up every time it eventually converges on the parameter values that fit pretty well with the agents behavior and so in this case you can just see this is just showing how the how the parameter value has changed from the prior values and this is probably a clearer depiction of this where for the negative image I started it at a value of one which would be negative in the model and the posterior was less for point value it went up and forbade it went up and in that case the direction in which those move was consistent with what the actual values were that I plugged into the simulation which is what you want to see to know that your models recoverable so so I can actually just yes well I mean it's just it's just the depending on the sign right like in in engineering a lot of times it is cast as maximizing yeah I know I agree you can think about this yes it's identical - yeah yeah right and that is one of those things that yeah Karl Karl could have been clearer about when you draw these chips but right yeah yeah yeah hmm sure no and that's a--that's a good point I am that I should have mentioned and didn't um that yeah I mean it's just the point is yes that it is converging to an optimal value so just to actually show you a little bit and the actual so here is so now if you know you guys will not look at it here is the actual this AAC underscore model underscore example so this is the full script that I just kind of went through with snapshots piece by piece you know so clear I'll close all I just set the random number generator to shuffle just so it doesn't produce the exact same thing every time I set my parameter values to particular violet values um this is just again exactly just piece by piece what I showed you um I don't think I really left anything out and one thing I guess worth mentioning is is that you can either specify policies in terms of U or V you as if you want policies to be shallow which means they only want one step ahead and V is in practice if you want the policies to be deep so the agent whoops saw ahead all the way up to the end of it ahead all the way till the end of the trial you can specify it as V even if your trial if it's only one step though which I which I am kind of like to do because if you specify it as use and it won't spit out the dopamine simulations but okay so then anyway I mean this is this is just basically giving it labels giving the plotting routine labels this checking thing is just a thing that helps you know you didn't mess up something you didn't write something inconsistent in your generative model before you run it and then you run it then this is the VB trial plot then this is just a couple other plots that you can generate the curls written plotting scripts for and then here is if I want to run more than one trial then I can just expand the MVP and that I made so it's replicated over several trials and then you can just run that whole thing again with the same function and then their plotting routines for looking at behavior across trials and you know so if I do this it's run one of these normally it's very fast but on this computer it's kind of slow so I probably won't run the group one but but here you go this is the s I mean this is basically the exact same thing is the one I showed you but and this is a another plot that I'm probably not gonna go through this it's not all that interesting in the context of this task but they're just it's just other depictions of things in the model or what the neural process theory would predict but if you want to know more about this kind of thing I can we can talk about it later it's just going to be more confusing for now I think but if if this if it wasn't so well maybe I can do it I was gonna say it's going to say it might take too long to run the multiple trials version but it's probably worth you guys seeing a little bit what it looks like so more or less this is just so this is showing five trials and these are just different sorts of plots that I'm more or less just encode the free-energy expected utility of the observations on each trial and they're just kin different kind of ways of coding the same thing is in the the single trial plots but just over multiple trials and again if you want to know details about what each of those mean then I can I can them we can talk about it individually later it's not that but important for now but in the and then in the second script that I gave you guys which is this inversion example script this has everything you need to actually do the parameter estimation model inversion that I that I showed you the example of where the thing will converge on the optimal for energy value it's also probably a little involved to go into in detail but here I just set the number of trials and then I set the parameter values that I to fixed values that I don't want to estimate then I set the some values that don't actually they're that well in this case I'm setting the values the true parameter values for the simulation I'm going to run I just kind of specify them here so that it will spit them out at the end so I can compare them to the actual model estimates and this is just putting together the U and why all this is just putting together the U and Y vectors that are going to be the observations and the actions that gets fed into the estimation routine and then down here more or less I just specify which of the parameters I actually want to estimate and then all of that goes into this DCM structure along with the U and Y which are the stimuli and the actions and then you run this and then you turn that into you take that DCM and you run it through the actual a it's the inversion script that I wrote which is just this function that's appended below and this is the one that in turn you set the priors for so this is basically saying if I want to I want to say that my prior for beta is one and I put that in log space to keep it positive so that during estimation it doesn't become a negative number because beta can't be a negative number and then same thing for point value and for any of these others but it will only catch them if I say I want to estimate these so in this case it'll estimate beta a negative image and point value and so here the priors that put R yes mm-hmm and yeah so it will so basically what I'll do is I just I just exponentiate it down here so it stays nonzero it's literally just a trick so it never becomes negative during the estimation but it so anyway so you do that and then at the end of the day it goes into this guy which is the actual estimation routine that runs the variational base and then you also need a likelihood function which is written here basically it just takes you start out with an L value of 0 and then it will repeatedly just add the previous likelihood value to the log probability of the basically the probability that the model assigns to the action that the agent actually chose and so it tries to maximize that over over iterations what do you say oh it's just it's just adding a little kind of small number to make sure that it never becomes 0 and then this is just reproducing the generative model that we already walked through but just as a function and then that's it so you can toy around with this obviously I've skipped through a little kind of just procedural things to just assigning different variables to other things just for sake of time but if I run this then it will simulate 60 trials and then it will try to fit them and I probably will take way too long to even show you any of it but give it a sec well that's that's just a single trial results yeah give it like another I don't know see it produced its first I likelihood estimate of negative 59 and it all so basically it'll it'll bounce around like small fluctuation around that number for a certain number of iterations basically until it's kind of tried out little adjustments to each of the parameters and then it'll jump on the next iteration to whatever goes down the gradient but I'm not going to not gonna run that the whole way because no way anytime so okay so anyway so that's so that's kind of beginning to end thoroughly through one task which Artie I realize it's probably a lot there were you know maybe we don't have to go into as much detail and the other ones but I mean another task that we were kind of excited about that's like an even kind of simpler model is this model that we built of an interceptive awareness task and this is meant to kind of capture this idea that a lot of people in computational psychiatry are now kind of like this idea that what might be going wrong is something about precision estimates assigned to afferent signals from the body and the way that contributes to either representing an understanding bodily States and also subsequent viscera motor regulation and so but no one's actually kind of try to explicitly show in any kind of model that different clinical groups actually do a sign look like they assign different precision estimates to after an intercept of signals and so we thought we'd give it a go and so we use this kind of really simple heart be tapping task and the idea is literally just for sixty seconds we just say tap a button whenever you feel your heart beat and then we simultaneously record EKGs and pulse transit time and things like that so we know when their actual heart beat was and what the soonest is that they would have felt it and so we did this into three conditions plus a control condition one where we said just guess do your best another where we said don't guess only press the button if you're totally sure I mean a third one where the no guessing stipulation was still in place but we also did an intercepted perturbation where we had them hold their breath with the hope of kind of amplifying the afferent signal and the reason that's important is because most people at rest have a really really terrible conscious interoception only about 35% of people in these tasks typically show greater than chance performance when they're trying to like indicate when they feel their heartbeat so to really get things off of floor values in a way that would be meaningful you probably have to do some kind of perturbation where you're getting the kind of variance that will be more interesting but you had to like control right for like something like they're guessing rate and also something like how a conservative or liberal they're being and so we did that again we did that in that same sample um but in this case we actually just did to try it we just decided to divide it up a little more fine grain so it's healthy people people with depression and people with anxiety people with anxiety and depression eating disorders we have like 18 people and then the substance use group so it's a pretty broad clinical group this is the way they set up the model it looks more complicated than it is more or less what we did is we just divided we took their actual cardiac events and we divided and then we added 200 milliseconds from false transit time and then you we just basically divided the epochs in half so that between heartbeats if you tapped in the first half of the time before the next heartbeat then that counted as tapping during systole otherwise it was tapping as a day under diastole and so that's it so the observations here were just assists elite or a diastole which was just based on their EKG political strands of time and then the other set of observations is just the conjunction and of whether they tapped and assisted ly didn't happen assistly tap the no delay no tap than that these are this is uh considered the afferent input that the brain is getting from their from the body so does their it's equivalent to like a visual stimulus but just coming up from their heart and and then this then is just basically encoding what they want right they're gonna prefer to tap when these assist aliy to tap and to observe that they tapped and observe assisted lee and not observe a no tap innocently etc right so just just true positives false positives true later guys from it's your positive sure negatives false positives false negatives and and and so what we fit here is then just um so these see values essentially how much they wanted to observe tapping and systole and the inverse for no tapping in systole and and the same thing for diastole and tapping versus no tapping and and it's a little bit hard to understand intuitively but what this ends up acting as is just a bias for tapping versus not having so the the higher the if you look at if you just take the PMT um value which is essentially how much they prefer to not get a false positive versus PT which is how much they what the magnitude is that they prefer not getting a false negative then again Jeff it just ends up this ends up just giving you essentially a metric of their bias against tapping and so just it's just kind of a measure of you know or the kind of person that likes to tap a lot or not very much it's kind of like I kind of treat conservatism about it as that um the other kind of thing that you can do and again this isn't initially all that intuitive but if you just take the average of those two so they're the average magnitude of their preference of the Preferences they have for both of those things then if that's a positive value then what that means is that what that means is that they actually had a strong preference to tap after a heartbeat whereas if that if that is negative then it means that they were actually an anticipate or they reliably tried to tap right before heart rate so you can kind of differentiate between anticipatory versus reactive strategies okay and and then but the most interesting one that we wanted to estimate was interoceptive precision so essentially the precision of the mapping from systole and diastole x' to felt heart beats or not heart beats and where the other where the controllable hidden state factor was choosing to tap or not tap right so the higher the IP value equals more precise and then I already explained this kind of tapping no tapping bias thing and so this is the kind of thing that we find so this is the bias against tapping measure and as you can see as you would as you would hope that this value is much higher in the no testing condition and the grethel condition and then in the guessing conditions people are a lot more conservative and when they choose to tap in these two which is exactly what you would expect this is just a Kanaka parameter validation kind of thing and the sensory precision is really high in the control tone condition sorry I didn't explain that basically the control condition is they just kept hearing a tone and they had to tap whenever they heard a tone and so precision estimates are really high for auditory precision in the tone condition and they're super super low you know except for some people to look like outliers any others and for the anticipate versus react thing the average of the see magnitudes what you get is basically a big chunk of people that are around zero which basically means they were really inconsistent and they're tapping which is just correlated with a sensory precision but then you have a good chunk of people that are on either end that are kind of reliable anticipators versus reliable reactors and so you can think some is like relying a lot more on prior expectations and others relying a lot on warrants than 3-input and so in terms of further parameter validation we just add we just looked at a bunch of different things so pulsed transit time a number of heartbeats a heart rate I'm for each person self-reported difficulty of the task self-reported confidence that they did it right self-reported felt heartbeat intensity and then age and BMI so body mass index and there's all things that influence cardiac perception and so what we found it's kind of cool is that for interoceptive precision and it was positively associated with confidence and intensity even in the guessing condition and whereas in the no guessing in the breath would condition both where this conservatism was higher the bias against tapping was associated in the directions you'd expect with difficulty and confidence in intensity and those are actually pretty similar and it's nothing else all that interesting what's kinda interesting even in the tone condition when that precision is essentially maximal you do get this kind of age difference in whether people are anticipated versus reactors which was kind of interesting and there weren't really all that many interesting differences or any interesting relationships here basically sensory precision in one condition didn't really map all that well the sensory precision and other conditions between the no guessing condition in the breath old condition there was more similarity and sensory precision but not that much biases against tapping was actually looked like it was kind of more trade ish across conditions for people and but the the interesting result is that the main interesting result is that when you look across groups for our interceptive precision and estimates they the healthy controls do have significantly higher interceptive precision than all of the clinical groups and the any other interesting effect that we saw was that actually medication status strongly blunted interceptive precision so if you were on any kind of psychiatric medication your interoceptive precision was much less and anyway doesn't look like we have time but I could have gone through basically how we specify the model for that and the only thing probably worth mentioning is the way that we specified interoceptive precision so in this case the way to fit this kind of precision parameter is you start just with diastole and systole mapping as an identity matrix to feeling a heartbeat and not but then you can pass this thing through a softmax function with a particular temperature parameter so it ends up happening is if this PA or intercept of precision is high it ends up looking like an identity matrix whereas the lower you may have to get this thing the more imprecise it ends up looking right so if person has values down here that means that the signal they get from systole and diastole provides a lot less precise evidence for the presence of a beat or note B and yes yeah I mean and again I can go through you know all this with you guys more you know just in person one of one at a time or whatever if you guys want to but these are you know I just kind of listed the whole bottle up here in case you guys want to look at it but the the way you estimate parameters is the same definitely won't have time to go into the UM band a reward learning version or our or our structure learning stuff but hopefully this was helpful and Casper's gonna have a bunch of time tomorrow to take us further so yeah any any questions or anything like that I mean obviously it would ask you something away you [Applause]